Vector Calculus: A lecture notes (under preparation)

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Be careful about typos

Contents

1 Vectors in space

1.1 Plane in space

1.1.1 Normal Form of the Equation of a Plane

Consider a plane in space and let O be the origin. Let N be the foot of the perpendicular drawn from O to the plane. Denote the length of this perpendicular by

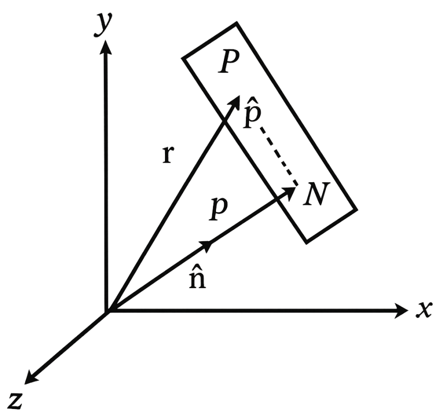
$$ON = p,$$

where p is the (perpendicular) distance of the plane from the origin.

Let $\hat{n}$ be a unit vector normal to the plane. Suppose

$$\hat{n} = (l, m, n),$$

where l, m, n are the direction cosines of the normal, so that



$$l^2 + m^2 + n^2 = 1.$$

Let $P(x, y, z)$ be any arbitrary point on the plane. Then the position vector of P is

$$\overrightarrow{OP} = \mathbf{r} = (x, y, z).$$

Since N lies on the plane and ON is perpendicular to the plane, the vector $\overrightarrow{ON}$ is along the normal direction $\hat{n}$. Hence,

$$\overrightarrow{ON} = p\,\hat{n}.$$

Now, the vector $\overrightarrow{NP} = \overrightarrow{OP} - \overrightarrow{ON}$ lies entirely in the plane. Therefore, it is perpendicular to the normal vector $\hat{n}$. Hence,

$$(\mathbf{r} - p\hat{n}) \cdot \hat{n} = 0.$$

Expanding, we get

$$\mathbf{r} \cdot \hat{n} - p(\hat{n} \cdot \hat{n}) = 0.$$

Since $\hat{n}$ is a unit vector, $\hat{n} \cdot \hat{n} = 1$, so

$$\mathbf{r} \cdot \hat{n} = p.$$

Substituting $\mathbf{r} = (x, y, z)$ and $\hat{n} = (l, m, n)$, we obtain

$$(x, y, z) \cdot (l, m, n) = lx + my + nz = p.$$

Thus, the **normal form of the equation of a plane** is

$$\boxed{\mathbf{r} \cdot \hat{n} = p} \quad \text{or} \quad \boxed{lx + my + nz = p}. \tag{1}$$

1.1.2 General Equation of a Plane

Starting from the normal form

$$\mathbf{r} \cdot \hat{n} = p,$$

we multiply both sides by a non-zero constant λ to obtain

$$\lambda(\mathbf{r} \cdot \hat{n}) = \lambda p.$$

Let us define a new vector

$$\mathbf{N} = \lambda \hat{n} = (a, b, c),$$

which is a (not necessarily unit) normal vector to the plane. Then the equation becomes

$$\mathbf{r} \cdot \mathbf{N} = \lambda p.$$

Let $d = -\lambda p$. Then the equation can be written as

$$\mathbf{r} \cdot \mathbf{N} + d = 0.$$

In Cartesian form, this becomes

$$ax + by + cz + d = 0.$$

This is called the **general equation of a plane**.

1.1.3 Equations of some special planes

1. **Equations of the planes passing through a given point:** Let the equation of plane passing through a given point with position vector $\mathbf{a}$ be

$$\mathbf{r} \cdot \mathbf{N} + d = 0 \tag{2}$$

   Since this plane passes through $\mathbf{a}$, we get $\mathbf{a} \cdot \mathbf{N} + d = 0 \Rightarrow d = -\mathbf{a} \cdot \mathbf{N}$. Putting the value of d in (2), we get

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{N} = 0 \tag{3}$$

2. **Equations of the coordinate planes**

   Since normal vector to the yz-plane is $\hat{\imath}$ and it passes through origin, we get

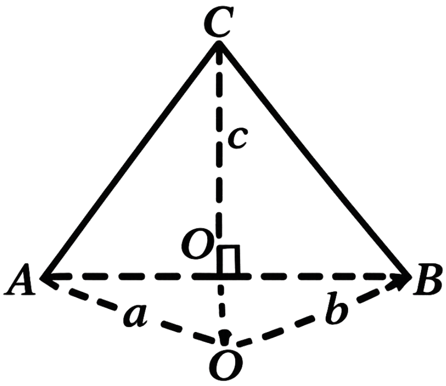
$$(\mathbf{r} - \mathbf{0}) \cdot \hat{\imath} = 0 \Rightarrow x = 0$$

   Similarly,

$$y = 0 \quad \text{and} \quad z = 0$$

3. **Equation of the plane passing through three non-collinear points:**

   Let the position vectors of three non-collinear points A, B, C be $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively.



Then a normal vector to the plane ABC is

$$\begin{aligned}\mathbf{N} = \overrightarrow{AB} \times \overrightarrow{AC} &= (\mathbf{b} - \mathbf{a}) \times (\mathbf{c} - \mathbf{a}) \\ &= \mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b}\end{aligned}$$

Hence, using the equation (3), the equation of the plane ABC is

$$\begin{aligned}(\mathbf{r} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b}) = 0 \\ \text{or} \quad \mathbf{r} \cdot (\mathbf{b} \times \mathbf{c} + \mathbf{c} \times \mathbf{a} + \mathbf{a} \times \mathbf{b}) = \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) \\ \text{or} \quad [\mathbf{r}\ \mathbf{b}\ \mathbf{c}] + [\mathbf{r}\ \mathbf{c}\ \mathbf{a}] + [\mathbf{r}\ \mathbf{a}\ \mathbf{b}] = [\mathbf{a}\ \mathbf{b}\ \mathbf{c}]\end{aligned}$$

1.2 Angle Between Two Planes

The angle between two planes is defined as the angle between their normals drawn through a point.

Let θ be the angle between two planes

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0 \quad \text{or} \quad a_1 x + b_1 y + c_1 z + d_1 = 0 \tag{4}$$

$$\text{and} \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0 \quad \text{or} \quad a_2 x + b_2 y + c_2 z + d_2 = 0 \tag{5}$$

Then the angle θ between their normal vectors $\mathbf{N}_1 = (a_1, b_1, c_1)$ and $\mathbf{N}_2 = (a_2, b_2, c_2)$ is given by

$$\cos\theta = \frac{\mathbf{N}_1 \cdot \mathbf{N}_2}{|\mathbf{N}_1||\mathbf{N}_2|} = \frac{a_1 a_2 + b_1 b_2 + c_1 c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2}\sqrt{a_2^2 + b_2^2 + c_2^2}}$$

1. **Condition of perpendicularity**

   Two planes (4) and (5) are perpendicular iff

$$\mathbf{N}_1 \cdot \mathbf{N}_2 = 0$$

   or

$$a_1 a_2 + b_1 b_2 + c_1 c_2 = 0$$

2. **Condition of parallelism**

   Two planes (4) and (5) are parallel iff $\mathbf{N}_1$ and $\mathbf{N}_2$ are collinear:

$$\mathbf{N}_2 = \lambda \mathbf{N}_1 \quad \text{or} \quad \frac{a_2}{a_1} = \frac{b_2}{b_1} = \frac{c_2}{c_1}$$

3. **Equation of a plane parallel to a given plane**

   Equation of a plane parallel to the plane

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0$$

   is

$$\mathbf{r} \cdot (\lambda \mathbf{N}_1) + d_2 = 0$$

   or

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1' = 0 \quad \text{where } d_1' = \frac{d_2}{\lambda}$$

   Thus, equations of parallel planes differ only by a constant.

1.3 Positions of Two Points Relative to a Plane

Theorem 1.3.1. *Two points A and B with position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively lie on the same or different sides of a plane $\mathbf{r} \cdot \mathbf{N} + d = 0$ according as the expressions $\mathbf{a} \cdot \mathbf{N} + d$ and $\mathbf{b} \cdot \mathbf{N} + d$ are of the same sign or different signs.*

Proof: Suppose that neither of the two points lies on the plane and that the line AB meets the plane at a point P which divides AB in the ratio $\lambda : 1$.

Then the position vector of P is

$$\frac{\mathbf{a} + \lambda \mathbf{b}}{1 + \lambda}$$

Since P lies on the plane,

$$\left(\frac{\mathbf{a} + \lambda \mathbf{b}}{1 + \lambda} \right) \cdot \mathbf{N} + d = 0$$

or

$$\lambda = -\frac{\mathbf{a} \cdot \mathbf{N} + d}{\mathbf{b} \cdot \mathbf{N} + d}$$

which shows that λ is positive or negative according as the expressions $\mathbf{a} \cdot \mathbf{N} + d$ and $\mathbf{b} \cdot \mathbf{N} + d$ have the same or different signs.

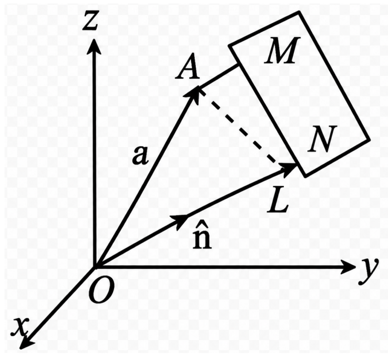
1.3.1 Distance Of a Point from a Plane

Let the position vector of a point A be $\mathbf{a}$ and the equation of the plane be

$$\mathbf{r} \cdot \hat{n} = p$$

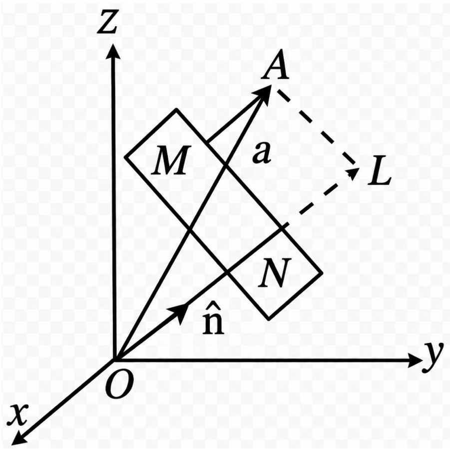
Case I: When the point A and the origin lie on the same side of the plane.

Perpendicular distance:



$$\text{Distance} = p - \mathbf{a} \cdot \hat{n}$$

Case II: When the point A and the origin lie on opposite sides of the plane.



$$\text{Distance} = \mathbf{a} \cdot \hat{n} - p$$

Hence, combining both cases, the perpendicular distance is

$$\boxed{|\mathbf{a} \cdot \hat{n} - p|}$$

$$\pm(p - \mathbf{a} \cdot \hat{n})$$

where the positive or negative sign is taken according as the point **a** and the origin lie on the same side of the plane or on the opposite sides.

Note: (i) By writing the general equation of the plane

$$\mathbf{r} \cdot \mathbf{N} + d = 0 \quad (d > 0)$$

in the normal form and using (1.3.1), we can obtain the perpendicular distance of a point **a** from this plane as

$$\pm \frac{\mathbf{a} \cdot \mathbf{N} + d}{|\mathbf{N}|}, \quad |\mathbf{N}| = N$$

(ii) The length of perpendicular from the point (x_1, y_1, z_1) to the plane

$$ax + by + cz + d = 0 \quad (d > 0)$$

is

$$\pm \frac{ax_1 + by_1 + cz_1 + d}{\sqrt{a^2 + b^2 + c^2}}$$

1.4 Planes Bisecting The Angles Between Two Planes

Let the equations of the intersecting planes be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad d_1 > 0$$

$$\mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0, \quad d_2 > 0$$

Then, by (1.3.1), the equations of the planes bisecting the angles (acute and obtuse) between the above planes are

$$\frac{\mathbf{r} \cdot \mathbf{N}_1 + d_1}{|\mathbf{N}_1|} = \pm \frac{\mathbf{r} \cdot \mathbf{N}_2 + d_2}{|\mathbf{N}_2|}$$

The positive sign gives the plane bisecting the angle between the given planes in which the origin lies; while the negative sign gives the plane bisecting the other angle.

1.4.1 Position Of the Origin Relative to Two Planes

Let the equations of the two intersecting planes be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad d_1 > 0$$

$$\mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0, \quad d_2 > 0$$

The normal forms of these equations are

$$\mathbf{r} \cdot \hat{n}_1 = p_1, \quad \mathbf{r} \cdot \hat{n}_2 = p_2$$

where

$$\hat{n}_1 = -\frac{\mathbf{N}_1}{|\mathbf{N}_1|}, \quad p_1 = \frac{d_1}{|\mathbf{N}_1|}, \quad |\mathbf{N}_1| = N_1$$

$$\hat{n}_2 = -\frac{\mathbf{N}_2}{|\mathbf{N}_2|}, \quad p_2 = \frac{d_2}{|\mathbf{N}_2|}, \quad |\mathbf{N}_2| = N_2$$

Clearly the origin lies in the acute or obtuse angle between the planes according as the angle θ between their normals is obtuse or acute, or equivalently, according as $\cos\theta$ is negative or positive.

Hence, the origin lies in the acute or obtuse angle between the planes according as

$$\hat{n}_1 \cdot \hat{n}_2$$

is negative or positive, or according as

$$\mathbf{N}_1 \cdot \mathbf{N}_2$$

is negative or positive.

Example 1.1 Find the equation of the plane through the point $(-1, 2, 3)$ and perpendicular to the line joining the points $(-4, -2, 5)$ and $(-2, 1, 1)$.
Solution. The plane passes through the point

$$\mathbf{a} = -\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$$

and is perpendicular to the vector $\overrightarrow{PQ}$.

Position vectors:

$$\begin{aligned}\mathbf{P} &= -4\hat{\imath} - 2\hat{\jmath} + 5\hat{k}, \quad \mathbf{Q} = -2\hat{\imath} + \hat{\jmath} + \hat{k} \\ \overrightarrow{PQ} &= \mathbf{Q} - \mathbf{P} = (-2+4)\hat{\imath} + (1+2)\hat{\jmath} + (1-5)\hat{k} = 2\hat{\imath} + 3\hat{\jmath} - 4\hat{k}\end{aligned}$$

Equation of plane:

$$\begin{aligned}(\mathbf{r} - \mathbf{a}) \cdot \overrightarrow{PQ} &= 0 \\ [x\hat{\imath} + y\hat{\jmath} + z\hat{k} - (-\hat{\imath} + 2\hat{\jmath} + 3\hat{k})] \cdot (2\hat{\imath} + 3\hat{\jmath} - 4\hat{k}) &= 0 \\ 2(x+1) + 3(y-2) - 4(z-3) &= 0 \\ 2x + 3y - 4z + 8 &= 0\end{aligned}$$

Example 1.2 Find the angle between the planes $2x - y + z = 6$ and $x + y + 2z = 7$.
Solution.

$$\mathbf{N}_1 = (2, -1, 1), \quad \mathbf{N}_2 = (1, 1, 2)$$

Let θ be the angle between the planes.

$$\begin{aligned}\cos\theta = \frac{\mathbf{N}_1 \cdot \mathbf{N}_2}{|\mathbf{N}_1||\mathbf{N}_2|} &= \frac{2(1) + (-1)(1) + 1(2)}{\sqrt{2^2 + (-1)^2 + 1^2}\sqrt{1^2 + 1^2 + 2^2}} \\ &= \frac{2 - 1 + 2}{\sqrt{6}\sqrt{6}} = \frac{3}{6} = \frac{1}{2} \\ \theta &= \cos^{-1}\left(\frac{1}{2}\right)\end{aligned}$$

$$\begin{gathered}\cos\theta = \frac{\mathbf{N}_1 \cdot \mathbf{N}_2}{|\mathbf{N}_1||\mathbf{N}_2|} = \frac{2 - 1 + 2}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \\ \theta = \frac{\pi}{3}\end{gathered}$$

Example 1.3 Find the equation of the plane passing through the line of intersection of the planes

$$x + 2y + 2z - 4 = 0, \quad 2x + y - z + 5 = 0$$

and perpendicular to the plane $4x + 5y - 3z = 8$.
Solution.

Equations of given planes:

$$\begin{aligned} \mathbf{r}\cdot\mathbf{N}_1 - 4 = 0, &\quad \mathbf{N}_1 = (1,2,2) \\ \mathbf{r}\cdot\mathbf{N}_2 + 5 = 0, &\quad \mathbf{N}_2 = (2,1,-1) \\ \mathbf{r}\cdot\mathbf{N}_3 = 8, &\quad \mathbf{N}_3 = (4,5,-3) \end{aligned} \tag{6}$$

Let required plane be:

$$\begin{aligned} &(\mathbf{r}\cdot\mathbf{N}_1 - 4) + \lambda(\mathbf{r}\cdot\mathbf{N}_2 + 5) = 0 \\ &\Rightarrow \mathbf{r}\cdot(\mathbf{N}_1 + \lambda\mathbf{N}_2) - 4 + 5\lambda = 0 \end{aligned} \tag{7}$$

Since required plane is perpendicular to (6):

$$(\mathbf{N}_1 + \lambda\mathbf{N}_2)\cdot\mathbf{N}_3 = 0$$

$$\lambda = -\frac{\mathbf{N}_1\cdot\mathbf{N}_3}{\mathbf{N}_2\cdot\mathbf{N}_3} = -\frac{4+10-6}{8+5+3} = -\frac{8}{16} = -\frac{1}{2}$$

Substitute in ():

$$\begin{aligned} \mathbf{r}\cdot(\mathbf{N}_1 - \tfrac{1}{2}\mathbf{N}_2) - 4 - \tfrac{5}{2} &= 0 \\ \mathbf{r}\cdot(2\mathbf{N}_1 - \mathbf{N}_2) - 13 &= 0 \\ 2\mathbf{N}_1 - \mathbf{N}_2 = (2,4,4) - (2,1,-1) &= (0,3,5) \\ 3y + 5z &= 13 \end{aligned}$$

Example 1.4 The plane $x - 2y + 3z = 0$ is rotated through a right angle about its line of intersection with the plane $2x + 3y - 4z - 5 = 0$. Find the equation of the plane in its new position.

Solution.

Given:

$$\mathbf{r}\cdot\mathbf{N}_1 = 0, \quad \mathbf{N}_1 = (1,-2,2)$$

$$\mathbf{r}\cdot\mathbf{N}_2 - 5 = 0, \quad \mathbf{N}_2 = (2,3,-4)$$

Required plane passes through line of intersection and is perpendicular to plane (i):

$$\mathbf{r}\cdot(\mathbf{N}_1 + \lambda\mathbf{N}_2) - 5\lambda = 0$$

Condition:

$$(\mathbf{N}_1 + \lambda\mathbf{N}_2)\cdot\mathbf{N}_1 = 0$$

$$\lambda = -\frac{\mathbf{N}_1^2}{\mathbf{N}_1\cdot\mathbf{N}_2} = -\frac{1+4+4}{2-6-12} = \frac{9}{16}$$

Substitute:

$$\mathbf{r}\cdot(8\mathbf{N}_1 + 7\mathbf{N}_2) - 35 = 0$$

$$(22x + 5y - 4z) = 35$$

1.5 Exercises

1. Find the equation of the plane which bisects the obtuse angle between the planes

$$5x + 12y - 13z = 0 \quad \text{and} \quad 3x + 4y - 5z + 1 = 0.$$

2. Find the equation of the plane passing through the intersection of the planes

$$2x + 5y - 5z = 6 \quad \text{and} \quad 2x + 7y - 8z = 7$$

and the point $(-1, 4, 3)$.

3. Find the locus of a point the sum of squares of whose distances from the planes

$$x + y + z = 0 \quad \text{and} \quad x - 2y + z = 0$$

is equal to the square of its distance from the plane $x - z = 0$.

4. A variable plane which remains at a constant distance p from the origin cuts the coordinate axes in A, B, C. Show that:

   (a) the locus of the centroid of triangle ABC is

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{p^2}$$

   (b) the locus of the centroid of the tetrahedron $OABC$ is

$$\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{16}{p^2}$$

2 Symmetrical Form of the Equation of a Line

The symmetrical form of the equation of a line is determined if we know the position vector of a point on the line and a vector parallel to the line.

2.1 A Parametric Form

Let $\mathbf{a}$ be the position vector of the point A through which the line is passing and $\mathbf{b}$ be a vector parallel to the line. Let P be any point with position vector $\mathbf{r}$ on the line. Then

Since $\overrightarrow{AP}$ and $\mathbf{b}$ are collinear,

$$\begin{aligned} \overrightarrow{AP} &= t\mathbf{b} \\ \Rightarrow \quad \mathbf{r} - \mathbf{a} &= t\mathbf{b} \\ \Rightarrow \quad \mathbf{r} &= \mathbf{a} + t\mathbf{b} \end{aligned} \tag{8}$$

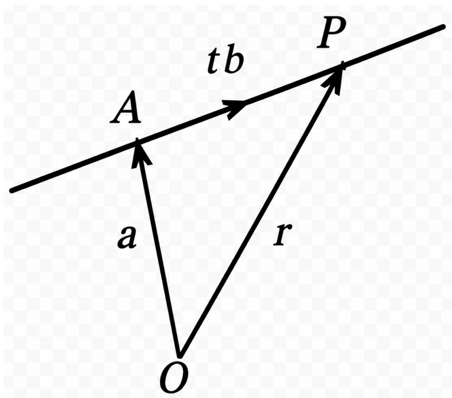
The equation (8) is called the parametric form of the vector equation of a line in symmetrical form.

2.1.1 Non-parametric Form

Taking cross product of (8) with $\mathbf{b}$, we obtain

$$(\mathbf{r} - \mathbf{a}) \times \mathbf{b} = \mathbf{0}$$

which is called the non-parametric form of the vector equation of a line.



2.1.2 Cartesian Equations

If (x, y, z) and (α, β, γ) are the coordinates of the points P and A respectively, with respect to an orthogonal frame of reference having origin O, and if $\mathbf{b} = (l, m, n)$, then the equation (8) may be written as

$$\frac{x-\alpha}{l} = \frac{y-\beta}{m} = \frac{z-\gamma}{n} \tag{9}$$

which are called the Cartesian equations of the symmetrical form of the equation of the line passing through the point (α, β, γ) and having direction ratios l, m, n.

2.2 Line through Two Points

Let the line pass through two points A and B whose position vectors are $\mathbf{a}$ and $\mathbf{b}$. Clearly $\overrightarrow{AB} = \mathbf{b} - \mathbf{a}$ is a vector parallel to the line. Its parametric vector equation, therefore, is

$$\mathbf{r} = \mathbf{a} + t(\mathbf{b} - \mathbf{a}) \tag{10}$$

If (x_1, y_1, z_1) and (x_2, y_2, z_2) are the coordinates of the points A and B respectively, then the equation (10) is equivalent to

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1} = \frac{z - z_1}{z_2 - z_1}$$

Note. If the line is regarded as passing through B and parallel to $\overrightarrow{AB}$, then its equation is

$$\mathbf{r} = \mathbf{b} + t(\mathbf{b} - \mathbf{a})$$

2.3 Non-Symmetrical Form of the Equation of a Line

The non-symmetrical form of the equation of a line are written from the consideration that a straight line is obtained by the intersection of two non-parallel planes. Therefore, it is completely determined when the equations of the planes are known.

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0 \tag{4.37}$$

are the equations of two planes passing through a straight line, then these two equations taken together represent the non-symmetrical form of equations of that line.

2.3.1 Non-parametric Form

Let the two non-parallel planes be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0 \tag{i}$$

Since the line of intersection of the planes (i) lies on both the planes, it is perpendicular to the normals of the two planes. Hence the line is parallel to the vector

$$\mathbf{N}_1 \times \mathbf{N}_2.$$

Let $\mathbf{a}$ be the position vector of a point on this line. Then $\mathbf{a}$ lies on the planes (i). Therefore,

$$\mathbf{a} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{a} \cdot \mathbf{N}_2 + d_2 = 0 \tag{ii}$$

Now, non-parametric form of the vector equation of the line passing through $\mathbf{a}$ and parallel to $\mathbf{N}_1 \times \mathbf{N}_2$ is

$$(\mathbf{r} - \mathbf{a}) \times (\mathbf{N}_1 \times \mathbf{N}_2) = \mathbf{0}$$

or

$$\mathbf{r} \times (\mathbf{N}_1 \times \mathbf{N}_2) = \mathbf{a} \times (\mathbf{N}_1 \times \mathbf{N}_2) = (\mathbf{a} \cdot \mathbf{N}_2)\mathbf{N}_1 - (\mathbf{a} \cdot \mathbf{N}_1)\mathbf{N}_2$$

Using (ii), this reduces to

$$\mathbf{r} \times (\mathbf{N}_1 \times \mathbf{N}_2) = d_1\mathbf{N}_2 - d_2\mathbf{N}_1$$

2.3.2 Transformation to Parametric Form

Let the non-symmetrical form of the equations of a line be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0$$

Since the line lies on both the planes, it is parallel to the vector $\mathbf{N}_1 \times \mathbf{N}_2$, and if a point with position vector $\mathbf{a}$ lies on this line, then by (8), the parametric form is

$$\mathbf{r} = \mathbf{a} + t(\mathbf{N}_1 \times \mathbf{N}_2)$$

2.4 Intersection of a Line and a Plane

In this section, we find the position vector of the point of intersection.

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0$$

are the equations of two planes passing through a straight line; then these two equations taken together represent the non-symmetrical form of the equations of that line.

2.4.1 A Non-parametric Form of the Vector Equation of the Line of Intersection of Two Non-parallel Planes

Let the two non-parallel planes be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0 \tag{i}$$

Since the line of intersection of the planes (i) lies on both the planes, it is perpendicular to the normals of the two planes. Hence the line is parallel to the vector

$$\mathbf{N}_1 \times \mathbf{N}_2.$$

Let $\mathbf{a}$ be the position vector of a point on this line. Then $\mathbf{a}$ lies on the planes (i). Therefore,

$$\mathbf{a} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{a} \cdot \mathbf{N}_2 + d_2 = 0 \tag{ii}$$

Now, non-parametric form of the vector equation of the line passing through $\mathbf{a}$ and parallel to $\mathbf{N}_1 \times \mathbf{N}_2$ is

$$(\mathbf{r} - \mathbf{a}) \times (\mathbf{N}_1 \times \mathbf{N}_2) = \mathbf{0}$$

or

$$\mathbf{r} \times (\mathbf{N}_1 \times \mathbf{N}_2) = \mathbf{a} \times (\mathbf{N}_1 \times \mathbf{N}_2) = (\mathbf{a} \cdot \mathbf{N}_2)\mathbf{N}_1 - (\mathbf{a} \cdot \mathbf{N}_1)\mathbf{N}_2$$

Using (ii), this reduces to

$$\mathbf{r} \times (\mathbf{N}_1 \times \mathbf{N}_2) = d_1\mathbf{N}_2 - d_2\mathbf{N}_1$$

2.5 Coplanar Lines (Intersecting Lines)

Let the equations of the straight lines be

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \tag{i}$$

and

$$\mathbf{r} = \mathbf{c} + s\mathbf{d} \tag{ii}$$

These lines, passing through $A(\mathbf{a})$ and $C(\mathbf{c})$, intersect at P if and only if their common plane is parallel to each of the vectors $\overrightarrow{CA}, \overrightarrow{PA}$ and $\overrightarrow{PC}$.

Hence the lines (i) and (ii) will intersect if the vectors $\overrightarrow{CA}, \overrightarrow{PA}$ and $\overrightarrow{PC}$ are coplanar.

$$\overrightarrow{CA} \cdot (\overrightarrow{PA} \times \overrightarrow{PC}) = 0$$

or

$$(\mathbf{a} - \mathbf{c}) \cdot (\lambda\mathbf{b} \times \mu\mathbf{d}) = 0$$

or

$$(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} \times \mathbf{d}) = 0$$

2.6 Planes Passing Through a Line

Case I: Symmetrical Form

Let the equation of the given line be

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}$$

which passes through the point $\mathbf{a}$ and is parallel to $\mathbf{b}$.

Since a plane passing through this line passes through $\mathbf{a}$ and is parallel to the line, the normal $\mathbf{N}$ to the plane is perpendicular to $\mathbf{b}$.

Consequently, the equation of a plane passing through the line is

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{N} = 0, \quad \mathbf{N} \cdot \mathbf{b} = 0$$

Case II: Non-symmetrical Form

Let the equation of the line in the non-symmetrical form be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{r} \cdot \mathbf{N}_2 + d_2 = 0$$

Then the equation of a plane passing through this line (i.e., the line of intersection of the planes) is

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 + \lambda(\mathbf{r} \cdot \mathbf{N}_2 + d_2) = 0$$

$$\mathbf{r} \cdot (\mathbf{N}_1 + \lambda\mathbf{N}_2) + d_1 + \lambda d_2 = 0$$

2.7 Plane Passing Through Two Coplanar Lines

Case I: Non-parallel Lines

Let the equations of two coplanar and non-parallel lines be

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \tag{i}$$

$$\mathbf{r} = \mathbf{c} + s\mathbf{d} \tag{ii}$$

Let the equation of a plane passing through line (i) be

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{N} = 0 \tag{iii}$$

where

$$\mathbf{N} \cdot \mathbf{b} = 0 \tag{iv}$$

Since this plane also passes through line (ii), the normal $\mathbf{N}$ is perpendicular to $\mathbf{d}$, i.e.

$$\mathbf{N} \cdot \mathbf{d} = 0 \tag{v}$$

From (iv) and (v), $\mathbf{N}$ is collinear with $\mathbf{b} \times \mathbf{d}$:

$$\mathbf{N} = \lambda(\mathbf{b} \times \mathbf{d})$$

Hence, the equation of the plane passing through both lines is

$$(\mathbf{r} - \mathbf{a}) \cdot (\mathbf{b} \times \mathbf{d}) = 0 \tag{11}$$

Case II: Parallel Lines

Let the equations of the lines be

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \tag{vi}$$

$$\mathbf{r} = \mathbf{c} + s\mathbf{b} \tag{vii}$$

Let the equation of a plane passing through (vi) be

$$(\mathbf{r} - \mathbf{a}) \cdot \mathbf{N} = 0 \tag{viii}$$

where

$$\mathbf{N} \cdot \mathbf{b} = 0 \tag{ix}$$

Since this plane also passes through (vii), it passes through $\mathbf{c}$:

$$(\mathbf{c} - \mathbf{a}) \cdot \mathbf{N} = 0 \tag{x}$$

From (ix) and (x), $\mathbf{N}$ is collinear with $\mathbf{b} \times (\mathbf{c} - \mathbf{a})$:

$$\mathbf{N} = \lambda\mathbf{b} \times (\mathbf{c} - \mathbf{a})$$

Hence, the equation of the plane passing through the parallel lines is

$$(\mathbf{r} - \mathbf{a}) \cdot \Big[\mathbf{b} \times (\mathbf{c} - \mathbf{a})\Big] = 0$$

2.8 Shortest Distance

Case I: When both lines are in symmetrical form

Let the equations of two non-intersecting lines be

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}$$

$$\mathbf{r} = \mathbf{c} + s\mathbf{d}$$

Let $PQ(= h)$ be the length of shortest distance (S.D.) between the lines. Since a vector along the line of shortest distance is perpendicular to both lines, $\mathbf{b} \times \mathbf{d}$ is a vector along it.

Then,

$$\text{S.D.} = h = OP = \text{projection of } \overrightarrow{CA} \text{ on the line of S.D.}$$

$$= \frac{|\overrightarrow{CA} \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|} = \frac{|(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|} \tag{12}$$

Case II: When the lines are in non-symmetrical form

Let the equations of the lines be

$$\mathbf{r} \cdot \mathbf{N}_1 + d_1 = 0 = \mathbf{r} \cdot \mathbf{N}_2 + d_2$$

$$\mathbf{r} \cdot \mathbf{N}_3 + d_3 = 0 = \mathbf{r} \cdot \mathbf{N}_4 + d_4$$

The given lines are parallel to the vectors $\mathbf{N}_1 \times \mathbf{N}_2$ and $\mathbf{N}_3 \times \mathbf{N}_4$ respectively.

Let the lines pass through the points $\mathbf{a}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} \cdot \mathbf{N}_1 + d_1 = 0, \quad \mathbf{a} \cdot \mathbf{N}_2 + d_2 = 0$$

$$\mathbf{c} \cdot \mathbf{N}_3 + d_3 = 0, \quad \mathbf{c} \cdot \mathbf{N}_4 + d_4 = 0$$

Then the equations of the lines are

$$\mathbf{r} = \mathbf{a} + t(\mathbf{N}_1 \times \mathbf{N}_2), \quad \mathbf{r} = \mathbf{c} + s(\mathbf{N}_3 \times \mathbf{N}_4)$$

Therefore, the shortest distance between these lines is

$$\text{S.D.} = \frac{|(\mathbf{a} - \mathbf{c}) \cdot \{(\mathbf{N}_1 \times \mathbf{N}_2) \times (\mathbf{N}_3 \times \mathbf{N}_4)\}|}{|(\mathbf{N}_1 \times \mathbf{N}_2) \times (\mathbf{N}_3 \times \mathbf{N}_4)|}$$

2.8.1 Equation of the Line of Shortest Distance

Case I: Non-symmetrical form

Clearly, the line of shortest distance between the lines is the line of intersection of the planes PQA and PQC.

As the plane PQA passes through two coplanar lines PA and PQ, according to (11), the equation of the plane PQA is

$$(\mathbf{r} - \mathbf{a}) \cdot \Big[\mathbf{b} \times (\mathbf{b} \times \mathbf{d})\Big] = 0 \tag{13}$$

Similarly, the equation of the plane PQC is

$$(\mathbf{r}-\mathbf{c})\cdot\Big[\mathbf{d}\times(\mathbf{b}\times\mathbf{d})\Big]=0 \tag{14}$$

Hence, the non-symmetrical form of the equations of the line of shortest distance between the lines

$$\mathbf{r}=\mathbf{a}+t\mathbf{b} \quad \text{and} \quad \mathbf{r}=\mathbf{c}+s\mathbf{d}$$

are given by (13) and (14) taken together.

Case II: Symmetrical form

Let the position vectors of the end points P and Q of the shortest distance be $\mathbf{a}+t\mathbf{b}$ and $\mathbf{c}+s\mathbf{d}$ respectively.

Then,

$$\overrightarrow{PQ}=(\mathbf{c}+s\mathbf{d})-(\mathbf{a}+t\mathbf{b})=(\mathbf{c}-\mathbf{a})+s\mathbf{d}-t\mathbf{b}$$

Since $\overrightarrow{PQ}$ is perpendicular to both the lines, we get

$$\overrightarrow{PQ}\cdot\mathbf{b}=0 \quad\Rightarrow\quad (\mathbf{b}\cdot\mathbf{d})s-b^2t=(\mathbf{a}-\mathbf{c})\cdot\mathbf{b} \tag{i}$$

and

$$\overrightarrow{PQ}\cdot\mathbf{d}=0 \quad\Rightarrow\quad d^2s-(\mathbf{b}\cdot\mathbf{d})t=(\mathbf{a}-\mathbf{c})\cdot\mathbf{d} \tag{ii}$$

Solving (i) and (ii), we get the values of s and t, and hence the position vectors of the points P and Q.

Consequently, the equation of the line joining P and Q, according to (4.35), is

$$\mathbf{r}=(\mathbf{a}+t\mathbf{b})+\lambda(\mathbf{c}-\mathbf{a}+s\mathbf{d}-t\mathbf{b}) \tag{4.55}$$

where s, t are given by (i), (ii) and λ is a scalar.

2.9 Angle Between a Line and a Plane

The angle between a line and a plane is defined as the complement of the angle between the line and the normal to the plane.

Let the equation of the line be

$$\mathbf{r}=\mathbf{a}+t\mathbf{b}$$

and that of the plane be

$$\mathbf{r}\cdot\mathbf{N}+d=0$$

Let θ be the angle between them. Then $90°-\theta$ is the angle between the line and the normal $\mathbf{N}$.

$$\cos(90°-\theta)=\frac{\mathbf{b}\cdot\mathbf{N}}{|\mathbf{b}||\mathbf{N}|}$$

$$\Rightarrow \sin\theta = \frac{\mathbf{b}\cdot\mathbf{N}}{|\mathbf{b}||\mathbf{N}|}$$

$$\theta = \sin^{-1}\left(\frac{\mathbf{b}\cdot\mathbf{N}}{|\mathbf{b}||\mathbf{N}|}\right) = \cos^{-1}\left(\frac{|\mathbf{b}\times\mathbf{N}|}{|\mathbf{b}||\mathbf{N}|}\right)$$

2.10 Volume of a Tetrahedron

Consider a tetrahedron $OABC$ whose concurrent edges $\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC}$ have the lengths and directions of the vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}$ respectively.

Then the volume of the tetrahedron is

$$V = \frac{1}{3}(\text{area of base triangle } OBC) \times (\text{height } AN)$$

$$= \frac{1}{3}\left(\frac{1}{2}|\mathbf{b}\times\mathbf{c}|\right)(|\mathbf{a}|\cos\theta)$$

$$= \frac{1}{6}|\mathbf{a}\cdot(\mathbf{b}\times\mathbf{c})|$$

Example 2.1 Find the equation of a plane which passes through a point $\mathbf{a}$ and is perpendicular to the plane $\mathbf{r}\cdot\mathbf{n} = q$ and parallel to the line $\mathbf{r} = \mathbf{b} + t\mathbf{c}$.

Solution. The equations of the given plane and line are

$$\mathbf{r}\cdot\mathbf{n} = q \tag{i}$$

$$\mathbf{r} = \mathbf{b} + t\mathbf{c} \tag{ii}$$

Let the equation of a plane passing through the point $\mathbf{a}$ be

$$(\mathbf{r}-\mathbf{a})\cdot\mathbf{N} = 0 \tag{iii}$$

Since this plane is perpendicular to plane (i) and parallel to line (ii),

$$\mathbf{N}\cdot\mathbf{n} = 0 \quad \text{and} \quad \mathbf{N}\cdot\mathbf{c} = 0$$

Thus $\mathbf{N}$ is collinear with $\mathbf{n}\times\mathbf{c}$:

$$\mathbf{N} = \lambda(\mathbf{n}\times\mathbf{c})$$

Hence the required plane is

$$(\mathbf{r}-\mathbf{a})\cdot(\mathbf{n}\times\mathbf{c}) = 0$$

Example 2.2 If $2d$ be the shortest distance between the lines

$$\frac{y}{b} + \frac{z}{c} = 1, \quad x = 0 \quad \text{and} \quad \frac{x}{a} - \frac{z}{c} = 1, \quad y = 0$$

prove that

$$\frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

Solution.

The given lines can be written as

$$\mathbf{r} = \mathbf{a} + t\mathbf{b}, \quad \mathbf{a} = (0, 0, c), \quad \mathbf{b} = \left(0, -\frac{1}{c}, \frac{1}{b}\right)$$

$$\mathbf{r} = \mathbf{c} + s\mathbf{d}, \quad \mathbf{c} = (a, 0, 0), \quad \mathbf{d} = \left(-\frac{1}{c}, 0, -\frac{1}{a}\right)$$

Shortest distance between lines:

$$2d = \frac{|(\mathbf{a} - \mathbf{c}) \cdot (\mathbf{b} \times \mathbf{d})|}{|\mathbf{b} \times \mathbf{d}|}$$

$$= \frac{|(-a, 0, c) \cdot \left(\frac{1}{ac}, \frac{1}{bc}, -\frac{1}{c^2}\right)|}{\sqrt{\frac{1}{a^2c^2} + \frac{1}{b^2c^2} + \frac{1}{c^4}}}$$

$$= \frac{\left|\frac{-a}{ac} + 0 + \frac{-c}{c^2}\right|}{\sqrt{\frac{1}{a^2c^2} + \frac{1}{b^2c^2} + \frac{1}{c^4}}}$$

Simplifying, we obtain

$$\frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

$$= \pm \frac{2}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}}$$

$$\therefore \quad \frac{1}{d^2} = \frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}$$

Example 2.3 Find the point of intersection of the lines

$$\mathbf{r} = \mathbf{a} + t\mathbf{b} \quad \text{and} \quad \mathbf{r} = \mathbf{c} + s\mathbf{d}$$

Solution.

Let $\mathbf{a} + \lambda\mathbf{b}$ be the point of intersection. Then

$$\mathbf{a} + \lambda\mathbf{b} = \mathbf{c} + s\mathbf{d}$$

Taking cross product with $\mathbf{d}$:

$$\mathbf{a} \times \mathbf{d} + \lambda(\mathbf{b} \times \mathbf{d}) = \mathbf{c} \times \mathbf{d}$$

$$\Rightarrow \lambda(\mathbf{b} \times \mathbf{d}) = (\mathbf{c} - \mathbf{a}) \times \mathbf{d}$$

$$\Rightarrow \lambda = \frac{|(\mathbf{c} - \mathbf{a}) \times \mathbf{d}|}{|\mathbf{b} \times \mathbf{d}|}$$

Hence the required point is

$$\mathbf{a} + \lambda\mathbf{b}$$

Example 2.4 Find the distance of the point $(1, -2, 3)$ from the plane $x - y + z = 5$ measured parallel to the line

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$$

Solution.

The given plane and line are

$$\mathbf{r} \cdot \mathbf{N} = 5, \quad \mathbf{N} = (1, -1, 1) \tag{i}$$

$$\mathbf{r} = t\mathbf{b}, \quad \mathbf{b} = (2, 3, -6) \tag{ii}$$

Let the equation of the line passing through the point $\mathbf{a} = (1, -2, 3)$ and parallel to (ii) be

$$\mathbf{r} = \mathbf{a} + \lambda\mathbf{b} \tag{iii}$$

The required distance is the distance between the given point and the point of intersection of (iii) and (i).

Let the point of intersection be $\mathbf{a} + \lambda\mathbf{b}$. Then

$$(\mathbf{a} + \lambda\mathbf{b}) \cdot \mathbf{N} = 5$$

$$\Rightarrow \lambda = \frac{5 - \mathbf{a} \cdot \mathbf{N}}{\mathbf{b} \cdot \mathbf{N}} = \frac{5 - (1 + 2 + 3)}{2 - 3 - 6} = \frac{-1}{-7} = \frac{1}{7}$$

Thus the point of intersection is

$$\mathbf{a} + \frac{1}{7}\mathbf{b}$$

Hence the required distance is

$$\left|\mathbf{a} + \frac{1}{7}\mathbf{b} - \mathbf{a}\right| = \left|\frac{1}{7}\mathbf{b}\right| = \frac{|\mathbf{b}|}{7} = \frac{7}{7} = 1$$

2.11 Exercises

1. Find the equation of the plane through the line

$$3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4$$

   and parallel to the line

$$x = 2y = 3z.$$

2. Find the distance of a point $\mathbf{a}$ from the plane $\mathbf{r} \cdot \mathbf{n} = q$ measured parallel to the line $\mathbf{r} = \mathbf{b} + t\mathbf{c}$.

3. Find the perpendicular distance of a corner of a unit cube from a diagonal not passing through it.

4. Find the length of the shortest distance between the lines $\mathbf{r} = \mathbf{a} + t\mathbf{b}$ and $\mathbf{r} = \mathbf{a} + s\mathbf{c}$ and determine the equation of the line which cuts both the lines at right angles.

5. Find the length and the equation of the shortest distance between the lines

$$\frac{x-2}{1} = \frac{y-1}{-2} = \frac{z-6}{1} \quad \text{and} \quad \frac{x+3}{7} = \frac{y+3}{-6} = \frac{z+3}{1}.$$

3 Inverse Function Theorem

The Inverse Function Theorem is a fundamental result in calculus and analysis. It gives conditions under which a function admits a locally defined inverse that is differentiable. It plays a central role in multivariable calculus, differential geometry, and optimization.

3.1 Inverse Function in One Variable

Definition 3.1.1. A function $f : I \subset \mathbb{R} \to \mathbb{R}$ is said to be invertible if there exists a function f^{-1} such that

$$f^{-1}(f(x)) = x \quad \text{for all } x \in I.$$

Theorem 3.1.2 (Inverse Function Theorem (One Variable))**.** *Let $f : (a, b) \to \mathbb{R}$ be continuously differentiable. If $f'(c) \neq 0$ for some $c \in (a, b)$, then there exists an interval around c on which f is invertible, and its inverse f^{-1} is differentiable.*

Moreover,

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}.$$

Remark 3.1.3. The condition $f'(c) \neq 0$ ensures that the function is locally monotonic and hence invertible.

3.1.1 Geometric Interpretation

If $f'(c) \neq 0$, the graph of f has a non-horizontal tangent at $x = c$. The inverse function is obtained by reflecting the graph about the line $y = x$.

3.2 Inverse Function Theorem in $\mathbb{R}^n$

Theorem 3.2.1 (Inverse Function Theorem (Multivariable))**.** *Let $F : \mathbb{R}^n \to \mathbb{R}^n$ be continuously differentiable in an open set containing a point $\mathbf{a}$. If the Jacobian matrix $DF(\mathbf{a})$ is invertible (i.e., $\det DF(\mathbf{a}) \neq 0$), then:*

- *There exists a neighborhood U of $\mathbf{a}$ and a neighborhood V of $F(\mathbf{a})$ such that $F : U \to V$ is bijective.*
- *The inverse function $F^{-1} : V \to U$ is continuously differentiable.*

Moreover,

$$D(F^{-1})(F(\mathbf{a})) = [DF(\mathbf{a})]^{-1}.$$

3.2.1 Jacobian Matrix

For $F = (f_1, f_2, \ldots, f_n)$, the Jacobian matrix is

$$DF = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix}$$

The condition $\det DF(\mathbf{a}) \neq 0$ ensures local invertibility.

3.3 Idea of Proof (Sketch)

3.3.1 One Variable

If $f'(c) \neq 0$, then f is strictly monotonic near c, hence invertible. Differentiating $f(f^{-1}(y)) = y$ gives:

$$f'(x) \cdot (f^{-1})'(y) = 1$$

3.3.2 Multivariable Case

The proof uses:

- Linear approximation via Jacobian
- Banach fixed-point theorem (in advanced treatments)

Example 3.1 Let $f(x) = x^3 + x$. Then

$$f'(x) = 3x^2 + 1 > 0 \quad \forall x$$

Hence f is invertible everywhere.

$$(f^{-1})'(y) = \frac{1}{3x^2 + 1}$$

where $x = f^{-1}(y)$.

Example 3.2 Let $F(x, y) = (x^2 - y^2, 2xy)$.

The Jacobian is:

$$DF = \begin{bmatrix} 2x & -2y \\ 2y & 2x \end{bmatrix}$$

$$\det DF = 4(x^2 + y^2)$$

Thus, F is locally invertible except at $(0, 0)$.

4 Implicit Function Theorem

The Implicit Function Theorem provides conditions under which a relation involving several variables defines one variable as a function of others. It is a fundamental tool in multivariable calculus, analysis, and differential geometry.

4.1 Implicit Functions

Definition 4.1.1. Let $f(x, y)$ be a function of two variables and $y = \varphi(x)$ be a function of x such that for every value of x for which $\varphi(x)$ is defined, $f(x, \varphi(x))$ vanishes identically (or in other words, $y = \varphi(x)$ is a solution of the functional equation $f(x, y) = 0$). Then $y = \varphi(x)$ is an implicit function defined by $f(x, y) = 0$.

In a similar way, we can define implicit function $z = \varphi(x, y)$ as a solution of $f(x, y, z) = 0$.

4.2 Implicit Function Theorem (One Equation)

Theorem 4.2.1 (Implicit Function Theorem)**.** *Let f be a real-valued continuous function defined on an open set $S(\subset \mathbb{R}^2)$. Let $(a, b) \in S$ and*

(i) $f(a, b) = 0$

(ii) f_x, f_y *are continuous in some neighbourhood of* (a, b)

(iii) $f_y(a, b) \neq 0$.

Then there exists a neighbourhood of (a, b), say $(a - h, a + h; b - k, b + k)(\subset S)$ such that for all $x \in (a - h, a + h)$, there exists one and only one $y = \varphi(x)$, $y \in (b - k, b + k)$, for which $f(x, y) = 0$ (identically), $x \in (a - h, a + h)$.

This function $y = \varphi(x)$ satisfies the following :

1. $b = \varphi(a)$

2. Both $\varphi(x)$ and $\varphi'(x)$ are continuous for each $x \in (a - h, a + h)$

3. $$\varphi'(x) = -\frac{f_x}{f_y}$$

4.2.1 Geometric Interpretation

The equation $f(x, y) = 0$ represents a curve. If $f_y \neq 0$, the curve can be locally written as $y = \varphi(x)$.

4.3 Implicit Function Theorem (General Form)

Theorem 4.3.1. *[Multivariable Implicit Function Theorem] Let $F : \mathbb{R}^{n+m} \to \mathbb{R}^m$ be continuously differentiable. Write variables as (x, y) where $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$.*

Suppose:

$$F(a, b) = 0$$

and the Jacobian matrix with respect to y,

$$D_y F(a, b),$$

is invertible (i.e., $\det D_y F(a, b) \neq 0$).

Then there exist neighborhoods U of a and V of b, and a unique differentiable function

$$y = g(x)$$

such that:

$$F(x, g(x)) = 0.$$

4.4 Derivative Formula

The derivative of the implicit function is given by:

$$Dg(x) = -[D_y F(x, g(x))]^{-1} D_x F(x, g(x))$$

4.5 Remarks

- The condition $\frac{\partial F}{\partial y} \neq 0$ is essential.
- The theorem guarantees *local* existence of functions.

4.6 Comparison with Inverse Function Theorem

- Inverse Function Theorem: solves $F(x) = y$
- Implicit Function Theorem: solves $F(x, y) = 0$

Example 4.1 Consider

$$F(x, y) = x^2 + y^2 - 1$$

Then

$$F_y = 2y$$

At $(0, 1)$, $F_y \neq 0$, so we can write y as a function of x locally.

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}$$

Example 4.2 Let

$$F(x, y, z) = x + y + z - 3$$

Then z can be expressed as:

$$z = 3 - x - y$$

Example 4.3 Consider the function

$$f(x, y) = x^2 + xy + y^2 - 1 = 0$$

in the neighbourhood of $(0, -1)$.

Here

$$f(0, -1) = 0, \qquad f_x = 2x + y, \qquad f_y = x + 2y$$

are continuous in the neighbourhood of $(0, -1)$ and

$$f_y(0, -1) = -2 \neq 0.$$

So it satisfies all the conditions stated in Theorem 4.3.1 on existence and uniqueness of implicit function in the neighbourhood of $(0, -1)$.

$$\begin{aligned} y^2 + xy + (x^2 - 1) = 0 & \\ \Rightarrow \quad y = \frac{-x \pm \sqrt{x^2 - 4(x^2 - 1)}}{2} & \\ \Rightarrow \quad y = \frac{1}{2}\left[-x \pm \sqrt{4 - 3x^2}\right], & \qquad |x| \leq \frac{2}{\sqrt{3}}. \end{aligned}$$

When $x = 0$, only

$$y = \frac{-x - \sqrt{4 - 3x^2}}{2}$$

gives $y = -1$.

Hence

$$y = \frac{1}{2}\left[-x - \sqrt{4 - 3x^2}\right]$$

is the required implicit function.

Example 4.4

$$f(x, y) = e^{x+y} + y - x = 0$$

in the neighbourhood of

$$\left(\frac{1}{2}, -\frac{1}{2}\right).$$

$$f\left(\frac{1}{2}, -\frac{1}{2}\right) = 0, \qquad f_x = e^{x+y} - 1, \qquad f_y = e^{x+y} + 1$$

are continuous in the neighbourhood of

$$\left(\frac{1}{2}, -\frac{1}{2}\right),$$

and

$$f_y\left(\frac{1}{2}, -\frac{1}{2}\right) = 2 \neq 0.$$

So there exists unique implicit function in neighbourhood of

$$\left(\frac{1}{2}, -\frac{1}{2}\right).$$

Example 4.5 The equation

$$\sin(x+y) + \sin(y+z) = 1$$

defines z implicitly as a function of x and y, say

$$z = f(x, y).$$

Compute the derivative

$$\frac{\partial^2 z}{\partial y\,\partial x}.$$

$$F(x, y, z) = \sin(x+y) + \sin(y+z) - 1 = 0$$

So

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{\cos(x+y)}{\cos(y+z)}, \qquad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{\cos(x+y)+\cos(y+z)}{\cos(y+z)}.$$

$$\frac{\partial^2 z}{\partial y\,\partial x} = -\frac{\partial}{\partial y}\left\{\frac{\cos(x+y)}{\cos(y+z)}\right\}$$

$$= -\frac{-\sin(x+y)\cos(y+z) + \cos(x+y)\sin(y+z)(z_y+1)}{\cos^2(y+z)}$$

$$= \frac{\sin(x+y)\cos^2(y+z) + \cos^2(x+y)\sin(y+z)}{\cos^3(y+z)}$$

Example 4.6 The equation

$$f\left(\frac{y}{x}, \frac{z}{x}\right) = 0$$

defines z implicitly as a function of x and y, say

$$z = g(x, y).$$

Assume g_x, g_y are continuous.

Assuming

$$D_2 f\left(\frac{y}{x}, \frac{z(x,y)}{x}\right) \neq 0,$$

show that $g(x, y)$ is homogeneous function of degree 1. [$D_2 f$ means p.d. of f w.r.t. the second argument.]

Let $\dfrac{y}{x} = p, \qquad \dfrac{z}{x} = q$. So $f(p,q) = 0$ where $f_q \neq 0$ by hypothesis.

$$f(p,q) = 0 \Rightarrow f_p\,dp + f_q\,dq = 0.$$

$$dp = d\left(\frac{y}{x}\right) = -\frac{y}{x^2}dx + \frac{1}{x}dy, \qquad dq = d\left(\frac{z}{x}\right) = -\frac{z}{x^2}dx + \frac{1}{x}dz.$$

Hence we get

$$\begin{aligned} f_p\left(-\frac{y}{x^2}dx + \frac{1}{x}dy\right) + f_q\left(-\frac{z}{x^2}dx + \frac{1}{x}dz\right) &= 0 \\ \Rightarrow dz = \left(\frac{y}{x}\cdot\frac{f_p}{f_q} + \frac{z}{x}\right)dx - \frac{f_p}{f_q}dy &= g_x\,dx + g_y\,dy \\ \Rightarrow \frac{\partial g}{\partial x} = \frac{y}{x}\cdot\frac{f_p}{f_q} + \frac{z}{x}, \qquad \frac{\partial g}{\partial y} &= -\frac{f_p}{f_q} \\ \Rightarrow x\frac{\partial g}{\partial x} + y\frac{\partial g}{\partial y} &= z = g(x,y). \end{aligned}$$

By converse of Euler's theorem, $g(x,y)$ is homogeneous function of degree 1 in x and y.

Example 4.7 Test for the existence of Implicit function(s) near the point indicated :

$$f(x,y) = \cos(x+y) + e^{y+x^2} + 3x - 2 - x^3y^3 = 0, \qquad (0,0).$$

We note that $f(0,0) = 0$. Here

$$\begin{aligned} f_x &= -\sin(x+y) + 2x\,e^{y+x^2} + 3 - 3x^2y^3 \\ f_y &= -\sin(x+y) + e^{y+x^2} - 3x^3y^2 \end{aligned}$$

and f_x and f_y are continuous functions or in other words, f is continuously differentiable. $f_y(0,0) = 1 \neq 0$ and so there exists implicit function $y = g(x)$ in neighbourhood of $(0,0)$ and $g'(0) = -\frac{f_x(0,0)}{f_y(0,0)} = -3$.

Also

$$f_x(0,0) \neq 0$$

and so there exists implicit function

$$x = h(y)$$

in neighbourhood of $(0,0)$ and

$$h'(0) = -\frac{f_y(0,0)}{f_x(0,0)} = -\frac{1}{3}.$$

Theorem 4.6.1. ***(Jacobian for Implicit Function)*** *If $u_1, u_2, \ldots, u_n$ instead of being given explicitly in terms of the variables $x_1, x_2, \ldots, x_n$, be connected with them by n equations*

$$F_i(u_1, u_2, \ldots, u_n, x_1, x_2, \ldots, x_n) = 0, \qquad (i = 1, 2, \ldots, n)$$

then

$$\frac{\partial(u_1,u_2,\ldots,u_n)}{\partial(x_1,x_2,\ldots,x_n)} = (-1)^n \frac{\dfrac{\partial(F_1,F_2,\ldots,F_n)}{\partial(x_1,x_2,\ldots,x_n)}}{\dfrac{\partial(F_1,F_2,\ldots,F_n)}{\partial(u_1,u_2,\ldots,u_n)}}.$$

Proof. By Chain rule,

$$\frac{\partial F_i}{\partial x_j} + \frac{\partial F_i}{\partial u_1}\frac{\partial u_1}{\partial x_j} + \frac{\partial F_i}{\partial u_2}\frac{\partial u_2}{\partial x_j} + \cdots + \frac{\partial F_i}{\partial u_n}\frac{\partial u_n}{\partial x_j} = 0 \qquad (1)$$

for all

$$i, j \in \{1, 2, \ldots, n\}.$$

$$\frac{\partial(F_1,F_2,\ldots,F_n)}{\partial(u_1,u_2,\ldots,u_n)}\frac{\partial(u_1,u_2,\ldots,u_n)}{\partial(x_1,x_2,\ldots,x_n)} = \begin{vmatrix} -\dfrac{\partial F_1}{\partial x_1} & -\dfrac{\partial F_1}{\partial x_2} & \cdots & -\dfrac{\partial F_1}{\partial x_n} \\ -\dfrac{\partial F_2}{\partial x_1} & -\dfrac{\partial F_2}{\partial x_2} & \cdots & -\dfrac{\partial F_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ -\dfrac{\partial F_n}{\partial x_1} & -\dfrac{\partial F_n}{\partial x_2} & \cdots & -\dfrac{\partial F_n}{\partial x_n} \end{vmatrix} \qquad \text{(using (1))}$$

$$= (-1)^n \frac{\partial(F_1,F_2,\ldots,F_n)}{\partial(x_1,x_2,\ldots,x_n)}.$$

Hence the result follows.

Theorem 4.6.2. *If $u = f(x,y)$ and $v = g(x,y)$ be such that all their first order partial derivatives are continuous and if*

$$\frac{\partial(u,v)}{\partial(x,y)} = 0,$$

then there exists a functional relation connecting u and v.

Proof.

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} u_x & u_y \\ v_x & v_y \end{vmatrix} = 0.$$

Case I : If v does not depend on y, then $\dfrac{\partial v}{\partial y} = 0$.

$$\Rightarrow \text{either } \frac{\partial u}{\partial y} = 0 \quad \text{or} \quad \frac{\partial v}{\partial x} = 0.$$

If $\dfrac{\partial v}{\partial x} = 0$, v is constant and the functional relation is $v = a \qquad (a \in \mathbb{R})$. If $\dfrac{\partial u}{\partial y} = 0$, u and v both are functions of x only. So $u = f(x), \qquad v = g(x)$.

Regarding x as a function of v and substituting in

$$u = f(x),$$

the functional relation will be obtained.
Case II. If v does depend on y,

$$\frac{\partial v}{\partial y} \neq 0.$$

The equation $v = g(x, y)$ defines y as a function of x and v, say $y = \psi(x, v)$ and putting it in $u = f(x, y)$, we get an equation of the form $u = F(x, v)$. Then by Chain rule,

$$\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial F}{\partial x} \cdot 1 + \frac{\partial F}{\partial v}\frac{\partial v}{\partial x} \\ \frac{\partial u}{\partial y} &= \frac{\partial F}{\partial v}\frac{\partial v}{\partial y} \end{aligned}$$

Hence

$$\begin{vmatrix} F_x + F_v\, v_x & F_v\, v_y \\ v_x & v_y \end{vmatrix} = 0$$
$$\Rightarrow \begin{vmatrix} F_x & 0 \\ v_x & v_y \end{vmatrix} = 0$$
$$\Rightarrow F_x = 0 \qquad \text{as} \qquad v_y \neq 0.$$

Hence we get $u = F(v)$ and the functional relation is obtained.

Theorem 4.6.3. *If $u_r = f_r(x_1, x_2, x_3)$ for $r = 1, 2, 3$ having continuous first order partial derivatives and*

$$\frac{\partial(u_1, u_2, u_3)}{\partial(x_1, x_2, x_3)} = 0,$$

then there exists a functional relation between u_1, u_2 and u_3.

Proof. u_3 must involve at least one of the variables x_3 (say), for if not the functional relation is $u_3 =$ constant.

So

$$\frac{\partial u_3}{\partial x_3} \neq 0$$

and we can express x_3 in terms of x_1, x_2 and u_3.

So for $r = 1, 2,$

$$u_r = g_r(x_1, x_2, u_3) = f_r\{x_1, x_2, x_3\} \tag{1}$$

By chain rule,

$$\begin{cases} \dfrac{\partial u_1}{\partial x_1} = \dfrac{\partial g_1}{\partial x_1} + \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_1}, \\ \dfrac{\partial u_1}{\partial x_2} = \dfrac{\partial g_1}{\partial x_2} + \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_2}, \\ \dfrac{\partial u_1}{\partial x_3} = \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_3}, \end{cases} \qquad \begin{cases} \dfrac{\partial u_2}{\partial x_1} = \dfrac{\partial g_2}{\partial x_1} + \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_1}, \\ \dfrac{\partial u_2}{\partial x_2} = \dfrac{\partial g_2}{\partial x_2} + \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_2}, \\ \dfrac{\partial u_2}{\partial x_3} = \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_3}. \end{cases}$$

Putting these in

$$\frac{\partial(u_1,u_2,u_3)}{\partial(x_1,x_2,x_3)} = 0,$$

$$\begin{vmatrix} \dfrac{\partial g_1}{\partial x_1} + \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_1} & \dfrac{\partial g_1}{\partial x_2} + \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_2} & \dfrac{\partial g_1}{\partial u_3}\dfrac{\partial u_3}{\partial x_3} \\ \dfrac{\partial g_2}{\partial x_1} + \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_1} & \dfrac{\partial g_2}{\partial x_2} + \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_2} & \dfrac{\partial g_2}{\partial u_3}\dfrac{\partial u_3}{\partial x_3} \\ \dfrac{\partial u_3}{\partial x_1} & \dfrac{\partial u_3}{\partial x_2} & \dfrac{\partial u_3}{\partial x_3} \end{vmatrix} = 0$$

$$\Rightarrow \frac{\partial(g_1,g_2)}{\partial(x_1,x_2)}\frac{\partial u_3}{\partial x_3} = 0$$

$$\Rightarrow \frac{\partial(g_1,g_2)}{\partial(x_1,x_2)} = 0 \qquad \text{as } \frac{\partial u_3}{\partial x_3} \neq 0$$

as g_1, g_2 are functionally related.

$$\Rightarrow u_1 \text{ and } u_2 \text{ are functionally related.}$$

By (1), u_1, u_2, u_3 are functionally related functions.

Theorem 4.6.4. *If u_1, u_2, u_3 be functions of x_1, x_2, x_3 having continuous first order partial derivatives and if there exists a functional relation between them, then*

$$\frac{\partial(u_1,u_2,u_3)}{\partial(x_1,x_2,x_3)} = 0.$$

Proof. Let the functional relation be

$$\phi(u_1,u_2,u_3) = 0.$$

So

$$d\phi = \frac{\partial \phi}{\partial u_1}du_1 + \frac{\partial \phi}{\partial u_2}du_2 + \frac{\partial \phi}{\partial u_3}du_3 = 0 \qquad (1)$$

Also

$$du_1 = \frac{\partial u_1}{\partial x_1}dx_1 + \frac{\partial u_1}{\partial x_2}dx_2 + \frac{\partial u_1}{\partial x_3}dx_3 \qquad (2)$$

$$du_2 = \frac{\partial u_2}{\partial x_1}dx_1 + \frac{\partial u_2}{\partial x_2}dx_2 + \frac{\partial u_2}{\partial x_3}dx_3 \qquad (3)$$

$$du_3 = \frac{\partial u_3}{\partial x_1}dx_1 + \frac{\partial u_3}{\partial x_2}dx_2 + \frac{\partial u_3}{\partial x_3}dx_3 \qquad (4)$$

By (1), (2), (3) and (4),

$$\sum_{r=1}^{3}\left(\frac{\partial\phi}{\partial u_1}\frac{\partial u_1}{\partial x_r}+\frac{\partial\phi}{\partial u_2}\frac{\partial u_2}{\partial x_r}+\frac{\partial\phi}{\partial u_3}\frac{\partial u_3}{\partial x_r}\right)dx_r=0.$$

As x_1, x_2, x_3 are independent,

$$\therefore \sum_{i=1}^{3}\left(\frac{\partial\phi}{\partial u_1}\frac{\partial u_1}{\partial x_t}+\frac{\partial\phi}{\partial u_2}\frac{\partial u_2}{\partial x_t}+\frac{\partial\phi}{\partial u_3}\frac{\partial u_3}{\partial x_t}\right)=0 \quad \text{for } t=1,2,3.$$

Eliminating

$$\frac{\partial\phi}{\partial u_1}, \qquad \frac{\partial\phi}{\partial u_2}, \qquad \frac{\partial\phi}{\partial u_3},$$

we have

$$\frac{\partial(u_1,u_2,u_3)}{\partial(x_1,x_2,x_3)}=0.$$

Example 4.8 Let

$$F(x,y,z,u)=0, \qquad G(x,y,z,u)=0, \qquad H(x,y,z,u)=0$$

where F, G, H have continuous first order partial derivatives with respect to all the variables. Is it always possible to have

(i) x, y and z in terms of u ?

Ans. Yes, if $\dfrac{\partial(F,G,H)}{\partial(x,y,z)}=0.$

(ii) x, y and u in terms of z ?

Ans. Yes, if $\dfrac{\partial(F,G,H)}{\partial(x,y,u)}=0.$

Hence consider the system of equations

$$\begin{aligned}3x+y-z-u^3&=0\\ x-y+2z+u&=0\\ 2x+2y-3z+2u&=0\end{aligned}$$

for (i) and (ii).

$$\left.\begin{aligned}F(x,y,z,u)&\equiv 3x+y-z-u^3=0\\ G(x,y,z,u)&\equiv x-y+2z+u=0\\ H(x,y,z,u)&\equiv 2x+2y-3z+2u=0\end{aligned}\right\}$$

$$\frac{\partial(F,G,H)}{\partial(x,y,z)} = \begin{vmatrix} 3 & 1 & -1 \\ 1 & -1 & 2 \\ 2 & 2 & -3 \end{vmatrix} = 0$$

So for (i), the answer is in the affirmative.

$$\frac{\partial(F,G,H)}{\partial(x,y,u)} = \begin{vmatrix} 3 & 1 & -3u^2 \\ 1 & -1 & 1 \\ 2 & 2 & 2 \end{vmatrix} = -12(u^2+1) \neq 0$$

So for (ii), the answer is in the negative.

Note also that

$$\begin{aligned} (3x+y-z)-(2x+2y-3z) &= x-y+2z \\ \Rightarrow u^3+2u &= -u \\ \Rightarrow u(u^2+3) &= 0 \qquad (\text{as } u \neq 0) \end{aligned}$$

4.7 exercises

1. Examine the applicability of the theorem on existence and uniqueness of Implicit function near the point indicated :

   (a)
   $$y^3 \cos x + y^2 \sin x = 4, \qquad \left(\frac{\pi}{2}, 2\right)$$

   (b)
   $$y^2 - 2xy + 5x^2 - 16 = 0, \qquad \left(1,\, 1-2\sqrt{3}\right)$$

   (c)
   $$x\cos(xy) = 0, \qquad \left(1, \frac{\pi}{2}\right)$$

   (d)
   $$xe^y - y + 1 = 0, \qquad (-1, 0)$$

   (e)
   $$y^3 + y - x^2 = 0, \qquad (0, 0)$$

2. Prove that if

$$f(x) = \int_0^x \frac{dt}{\sqrt{1-t^4}}, \qquad 0 \le x < 1,$$

then without using the method of integration, show that

$$f(x) + f(y) = f\left\{\frac{x\sqrt{1-y^4} + y\sqrt{1-x^4}}{1+x^2y^2}\right\}.$$

3. Examine whether f, g, h are functionally related

$$\begin{aligned} f(x,y,z,w) &= x+y+z+w \\ g(x,y,z,w) &= x^2+y^2+z^2+w^2 \\ h(x,y,z,w) &= 2(xy+xz+xw+yz+yw+zw) \end{aligned}$$

4.

$$u = f(v,w,x), \qquad v = g(w,u,x), \qquad w = h(u,v,x)$$

all have continuous first order partial derivatives. Show that

$$\frac{du}{dx} = -\frac{\begin{vmatrix} f_x & f_v & f_w \\ g_x & -1 & g_w \\ h_x & h_v & -1 \end{vmatrix}}{\begin{vmatrix} -1 & f_v & f_w \\ g_u & -1 & g_w \\ h_u & h_v & -1 \end{vmatrix}}$$

5. Given the transformation

$$x = f(u,v,w), \qquad y = g(u,v,w), \qquad z = h(u,v,w)$$

with jacobian

$$J = \frac{\partial(x,y,z)}{\partial(u,v,w)} \ne 0,$$

show that for the inverse functions

$$\frac{\partial u}{\partial x} = \frac{1}{J}\frac{\partial(y,z)}{\partial(v,w)}, \qquad \frac{\partial u}{\partial y} = \frac{1}{J}\frac{\partial(z,x)}{\partial(v,w)}, \qquad \frac{\partial u}{\partial z} = \frac{1}{J}\frac{\partial(x,y)}{\partial(v,w)}.$$

5 Curves Defined by Parametric Equations

In many applications, curves are described not as $y = f(x)$ but by expressing both x and y in terms of a third variable called a *parameter*. Such representations are called parametric equations.

Definition 5.0.1. A plane curve is said to be defined parametrically if

$$x = x(t), \quad y = y(t), \quad t \in I$$

where t is a parameter and I is an interval.

5.1 Elimination of Parameter

If possible, eliminating t between $x(t)$ and $y(t)$ gives the Cartesian equation of the curve.

Example 5.1

$$x = t^2, \quad y = t$$

Eliminating t, we get $x = y^2$, a parabola.

Example 5.2

$$x = a\cos t, \quad y = a\sin t$$

This represents a circle of radius a.

5.2 Derivatives

5.2.1 First Derivative

If $x(t)$ and $y(t)$ are differentiable and $\frac{dx}{dt} \neq 0$, then

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$$

5.2.2 Second Derivative

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right) \bigg/ \frac{dx}{dt}$$

5.3 Tangents and Normals

Slope of Tangent

$$\frac{dy}{dx} = \frac{y'(t)}{x'(t)}$$

Equation of Tangent

At $t = t_0$:

$$y - y(t_0) = \frac{y'(t_0)}{x'(t_0)}(x - x(t_0))$$

5.4 Equation of Normal

$$y - y(t_0) = -\frac{x'(t_0)}{y'(t_0)}(x - x(t_0))$$

5.5 Arc Length

The length of a parametric curve from $t = a$ to $t = b$ is:

$$s = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

5.6 Area Under a Parametric Curve

The area under the curve is given by:

$$A = \int y\, dx = \int_a^b y(t)\frac{dx}{dt}\, dt$$

5.7 Area Enclosed by a Curve

$$A = \int_a^b x(t)\frac{dy}{dt}\, dt$$

5.8 Velocity and Acceleration

Let

$$\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j}$$

Velocity:

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt}$$

Speed:

$$|\mathbf{v}(t)| = \sqrt{(x'(t))^2 + (y'(t))^2}$$

Acceleration:

$$\mathbf{a}(t) = \frac{d^2\mathbf{r}}{dt^2}$$

5.9 Concavity

- $\dfrac{d^2y}{dx^2} > 0$: Concave upward
- $\dfrac{d^2y}{dx^2} < 0$: Concave downward

5.10 Curvature

Definition 5.10.1. Curvature is defined as the rate of change of the direction of a curve as you move along it. Let $\tan\psi$ be the slope of the curve at point P. Then the curvature at point P is defined as the magnitude of rate of change of ψ with respect to the arc length s. Thus curvature at P is $\kappa = \left|\frac{d\psi}{ds}\right|$.

Radius of curvature is

$$\rho = \frac{1}{\kappa}.$$

We know that $\frac{dy}{dx} = \tan\psi$ and $ds^2 = dx^2 + dy^2$ i.e. $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$. Now

$$\begin{aligned}
\frac{dy}{dx} &= \tan\psi \\
\frac{d^2y}{dx^2} &= \sec^2\psi \frac{d\psi}{dx} \\
&= (1 + \tan^2 x)\frac{d\psi}{ds}\frac{ds}{dx} \\
&= \left(1 + \left(\frac{dy}{dx}\right)^2\right)\frac{d\psi}{ds}\sqrt{1 + \left(\frac{dy}{dx}\right)^2} \\
\Rightarrow \quad \kappa = \frac{d\psi}{ds} &= \frac{\frac{d^2y}{dx^2}}{\left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2}}.
\end{aligned}$$

Example 5.3 Find the radius of curvature for

$$\sqrt{\frac{x}{a}} + \sqrt{\frac{y}{b}} = 1$$

at the points where it touches the coordinate axes.

Solution :

On differentiating the given, we get

$$\begin{aligned}
\frac{1}{2\sqrt{ax}} + \frac{1}{2\sqrt{by}}\frac{dy}{dx} &= 0 \\
\Rightarrow \frac{dy}{dx} &= -\sqrt{\frac{by}{ax}} \qquad \cdots (1)
\end{aligned}$$

The curve touches the x-axis if $\frac{dy}{dx} = 0$ i.e, $y = 0$. When $y = 0$, we have $x = a$ (from the given equation). Thus, given curve touches x-axis at $(a, 0)$.

The curve touches the y-axis if $\frac{dx}{dy} = 0$ i.e. $x = 0$.

When $x = 0$, we have $y = b$

$\Rightarrow$ Given curve touches y-axis at $(0, b)$.

$$\frac{d^2y}{dx^2} = \sqrt{\frac{b}{a}} \left(\sqrt{\frac{b}{a}} \cdot \frac{1}{2x} \cdot \frac{1}{2\sqrt{\frac{y}{x}}} \right) \qquad \{\text{from (1)}\}$$

$$\text{At} \quad (a, 0), \qquad \frac{d^2y}{dx^2} = \frac{1}{2a} \cdot \frac{b}{a} = \frac{b}{2a^2}$$

$$\therefore \text{ At } (a, 0), \qquad \rho = \frac{(1 + y_1^2)^{3/2}}{y_2} = \frac{(1 + 0)^{3/2}}{\frac{b}{2a^2}} = \frac{2a^2}{b}$$

$$\text{At} \quad (0, b), \qquad \rho = \frac{\left[1 + \left(\frac{dx}{dy}\right)^2\right]^{3/2}}{\frac{d^2x}{dy^2}} = \frac{2b^2}{a}.$$

5.10.1 Curvature of parametric curve

Let the parametric form of the curve be

$$x = f(t) \quad \text{and} \quad y = g(t).$$

For the parametric curve,

$$ds^2 = dx^2 + dy^2.$$

Dividing by dt^2,

$$\left(\frac{ds}{dt}\right)^2 = \left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2.$$

Therefore,

$$\frac{ds}{dt} = \sqrt{x'^2 + y'^2},$$

where

$$x' = \frac{dx}{dt}, \qquad y' = \frac{dy}{dt}.$$

Tangent Angle:

The slope of the tangent is

$$\tan\psi = \frac{dy/dt}{dx/dt} = \frac{y'}{x'}.$$

Differentiate with respect to t:

$$\sec^2\psi \, \frac{d\psi}{dt} = \frac{x'y'' - y'x''}{x'^2}.$$

Now,

$$\sec^2\psi = 1 + \tan^2\psi = 1 + \left(\frac{y'}{x'}\right)^2 = \frac{x'^2 + y'^2}{x'^2}.$$

Substituting,

$$\frac{x'^2 + y'^2}{x'^2}\frac{d\psi}{dt} = \frac{x'y'' - y'x''}{x'^2}.$$

Hence,

$$\frac{d\psi}{dt} = \frac{x'y'' - y'x''}{x'^2 + y'^2}.$$

Formula for Curvature

Using

$$\kappa = \left|\frac{d\psi/dt}{ds/dt}\right| = \left|\frac{\dfrac{x'y'' - y'x''}{x'^2 + y'^2}}{\sqrt{x'^2 + y'^2}}\right| = \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}}$$
$$= \frac{|x'y'' - y'x''|}{(x'^2 + y'^2)^{3/2}}.$$

Curvature of a parametric curve is given by:

$$\kappa = \frac{|x'(t)y''(t) - y'(t)x''(t)|}{[(x'(t))^2 + (y'(t))^2]^{3/2}}$$

Radius of Curvature

Since

$$\rho = \frac{1}{\kappa},$$

we obtain

$$\boxed{\rho = \frac{(x'^2 + y'^2)^{3/2}}{|x'y'' - y'x''|}}$$

The final formula is:

$$\rho = \frac{(x'^2 + y'^2)^{3/2}}{|x'y'' - y'x''|}.$$

where

$$x' = \frac{dx}{dt}, \quad y' = \frac{dy}{dt}, \quad x'' = \frac{d^2x}{dt^2}, \quad y'' = \frac{d^2y}{dt^2}.$$

Example 5.4 Show that the radius of curvature at any point of the curve

$$x^{2/3} + y^{2/3} = a^{2/3}$$

$$(x = a\cos^3\theta,\ y = a\sin^3\theta)$$

is equal to three times the length of the perpendicular from the origin to the tangent.

Solution :

$$
\begin{aligned}
\frac{dx}{d\theta} &= -3a\cos^2\theta\sin\theta = x' \\
\frac{dy}{d\theta} &= 3a\sin^2\theta\cos\theta = y' \\
x'' &= \frac{d^2x}{d\theta^2} = \frac{d}{d\theta}\left(-3a\cos^2\theta\sin\theta\right) = -3a\left[-2\cos\theta\sin^2\theta + \cos^3\theta\right] \\
&= 6a\cos\theta\sin^2\theta - 3a\cos^3\theta \\
y'' &= \frac{d^2y}{d\theta^2} = \frac{d}{d\theta}\left(3a\sin^2\theta\cos\theta\right) = 3a\left(2\sin\theta\cos^2\theta - \sin^3\theta\right) \\
&= 6a\sin\theta\cos^2\theta - 3a\sin^3\theta.
\end{aligned}
$$

Now

$$
\begin{aligned}
\rho &= \frac{(x'^2 + y'^2)^{3/2}}{|x'y'' - y'x''|} \\
&= \frac{(9a^2\cos^4\theta\sin^2\theta + 9a^2\sin^4\theta\cos^2\theta)^{3/2}}{|(-3a\cos^2\theta\sin\theta)(6a\sin\theta\cos^2\theta - 3a\sin^3\theta) - 3a\sin^2\theta\cos\theta(6a\cos\theta\sin^2\theta - 3a\cos^3\theta)|} \\
&= \frac{[9a^2\cos^2\theta\sin^2\theta(\cos^2\theta + \sin^2\theta)]^{3/2}}{|-18a^2\sin^2\theta\cos^4\theta + 9a^2\cos^2\theta\sin^4\theta - 18a^2\sin^4\theta\cos^2\theta + 9a^2\sin^2\theta\cos^4\theta|} \\
&= \frac{9^{3/2}(a\cos\theta\sin\theta)^3}{|-9a^2\sin^2\theta\cos^4\theta - 9a^2\cos^2\theta\sin^4\theta|} \\
&= \frac{(9)^{3/2}(a\cos\theta\sin\theta)^3}{9a^2\cos^2\theta\sin^2\theta(\cos^2\theta + \sin^2\theta)}
\end{aligned}
$$

$$\Rightarrow \quad \rho = 3a\sin\theta\cos\theta \qquad \cdots (1)$$

The equation of the tangent at any point on the curve is

$$
\begin{aligned}
y - a\sin^3\theta &= -\tan\theta(x - a\cos^3\theta) \\
\Rightarrow x\sin\theta + y\cos\theta - a\sin\theta\cos\theta &= 0 \qquad \cdots (2)
\end{aligned}
$$

$\therefore$ The length of the perpendicular from the origin to the tangent (2) is

$$
\begin{aligned}
p &= \frac{|a\sin\theta\cos\theta|}{\sqrt{\sin^2\theta + \cos^2\theta}} \\
&= a\sin\theta\cos\theta \qquad \cdots (3)
\end{aligned}
$$

Hence from (1) & (3),

$$\rho = 3p$$

5.10.2 Radius of curvature of Polar curves $r = f(\theta)$:

$$\rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2} \qquad \left(\text{where } r_1 = \frac{dr}{d\theta},\ r_2 = \frac{d^2r}{d\theta^2}\right)$$

Example 5.5 Prove that for the cardioide $r = a(1 + \cos\theta)$, $\dfrac{\rho^2}{r}$ is constant.

Solution :

Here

$$\begin{aligned}
r &= a(1 + \cos\theta) \\
\Rightarrow r_1 &= -a\sin\theta \qquad \text{and} \qquad r_2 = -a\cos\theta \\
\therefore \quad r^2 + r_1^2 &= a^2\left[(1 + \cos\theta)^2 + \sin^2\theta\right] \\
&= 2a^2(1 + \cos\theta) \\
r^2 + 2r_1^2 - rr_2 &= a^2\left[(1 + \cos\theta)^2 + 2\sin^2\theta + \cos\theta(1 + \cos\theta)\right] \\
&= 3a^2(1 + \cos\theta) \\
\therefore \quad \rho^2 &= \frac{(r^2 + r_1^2)^3}{(r^2 + 2r_1^2 - rr_2)^2} \\
&= \frac{8a^6(1 + \cos\theta)^3}{9a^4(1 + \cos\theta)^2} = \frac{8}{9}a^2(1 + \cos\theta) \\
\Rightarrow \quad \rho^2 &= \frac{8a}{9}r \\
\therefore \quad \frac{\rho^2}{r} &= \frac{8a}{9},
\end{aligned}$$

which is a constant.

Example 5.6 Find the radius of curvature at any point (r, θ) of the curve

$$r^m = a^m \cos m\theta$$

Solution :

$$\begin{aligned}
r^m &= a^m \cos m\theta \\
\Rightarrow m\log r &= m\log a + \log(\cos m\theta) \\
\Rightarrow \frac{m}{r}r_1 &= -m\frac{\sin m\theta}{\cos m\theta} \qquad \text{(on differentiating w.r.t. } \theta) \\
\Rightarrow r_1 &= -r\tan m\theta \qquad \cdots (1) \\
\text{Now} \quad r_2 &= -\left(r_1\tan m\theta + rm\sec^2 m\theta\right) \\
&= r\tan^2 m\theta - rm\sec^2 m\theta \qquad \text{(from (1))}
\end{aligned}$$

$$\begin{aligned}
\therefore \quad \rho &= \frac{(r^2 + r^2\tan^2 m\theta)^{3/2}}{r^2 + 2r^2\tan^2 m\theta - r^2\tan^2 m\theta + r^2 m\sec^2 m\theta} \\
&= \frac{r^3\sec^3 m\theta}{r^2\sec^2 m\theta + r^2 m\sec^2 m\theta} \\
&= \frac{r}{m+1}\sec m\theta
\end{aligned}$$

5.11 Area Under a Parametric Curve

Area between the curve and the x-axis:

$$A = \int y\,dx = \int y(t)\frac{dx}{dt}dt$$

5.12 Symmetry

- Symmetric about x-axis if replacing t by $-t$ changes y to $-y$
- Symmetric about y-axis if replacing t by $-t$ leaves y unchanged and changes x to $-x$

5.13 Common Parametric Curves

5.13.1 Circle

$$x = a\cos t, \quad y = a\sin t$$

5.13.2 Cycloid

$$x = a(t - \sin t), \quad y = a(1 - \cos t)$$

5.13.3 Ellipse

$$x = a\cos t, \quad y = b\sin t$$

Example 5.7 Find $\frac{dy}{dx}$ for $x = t^2$, $y = t^3$.

$$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}$$

Example 5.8 Find the arc length of $x = t$, $y = t^2$ from $t = 0$ to $t = 1$.

$$s = \int_0^1 \sqrt{1 + (2t)^2}\,dt$$

5.14 Exercises

1. Find the radius of curvature of

$$y^2 = x^2(a+x)(a-x)$$

at the origin.

2. Find the radius of curvature at any point t of the curve

$$x = a\left(\cos t + \log\tan\frac{t}{2}\right), \qquad y = a\sin t$$

3. Find the radius of curvature to the curve

$$r = a(1+\cos\theta)$$

at the point where the tangent is parallel to the initial line.

4. For the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1,$$

prove that

$$\rho = \frac{a^2b^2}{p^3}$$

where p is the perpendicular distance from the centre to the tangent at (x, y).

5.15 References

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